

URA Aggregator (MMIS as an Aggregator + Using EFRIS)

URA EFRIS Aggregator / S2S Integrator for MMIS – Complete Guidance

To act as a **centralized aggregator** for vendors in the Uganda Market Master – MMIS (submitting e-invoices/receipts on their behalf to URA), you must officially become a **certified System-to-System (S2S) Integrator** with URA. Direct EFRIS integration does **not** avoid the aggregator route — the S2S API path *is* the integrator/aggregator route. URA treats platforms like MMIS as “integrators” when they handle submissions centrally for multiple taxpayers.

What is a URA EFRIS Aggregator / S2S Integrator?

- URA allows **third-party software platforms** (ERPs, market systems, billing apps) to connect directly to EFRIS via API.
- As an **integrator/aggregator**, MMIS can:
 - Validate TINs.
 - Generate and submit e-invoices/receipts on behalf of many vendors.
 - Handle PRN generation and daily batch remittance.
- This is the **recommended model** for high-volume ecosystems like MMIS (21 cities, thousands of vendors).

Individual vendors can still use the URA web portal manually, but MMIS loses control and compliance value.

How to Become a URA EFRIS Aggregator / S2S Integrator

URA’s official process (based on current EFRIS S2S Guide and integrator registration flow):

1. Register on URA EFRIS Portal (Test Environment)

- Go to the URA EFRIS portal → “System Integration” or “Software Integrator” section.
- Create an EFRIS account as a **Software Integrator / Service Provider**.
- Submit company details, TIN, and proof of business.

2. Download Technical Documentation

- URA provides:
 - EFRIS Interface User Requirement Specifications.
 - S2S Integration Guide (v1.1+).
 - API endpoints, encryption specs, and certificate requirements.

3. Apply for Integrator Approval

- Upload:

- Company registration documents.
- Sample integration architecture.
- Security & data protection policy (Uganda Data Protection Act compliance).
- URA reviews and approves you as a certified integrator.

4. **Technical Setup (S2S Integration)**

- **Digital Certificates:** Obtain public/private keys from a URA-approved Certification Authority. Rotate certificates as per URA policy (critical error handling in your v5.2 schema already covers this).
- **API Integration:**
 - URA's S2S endpoints (T101 server time, T102 key exchange, T103 invoice submit, etc.).
 - Encrypt payloads with AES + digital signatures.
 - Handle rate limits, 429 errors, TIN validation, and 5xx retries (your existing error handling strategy matches URA requirements).
- Test in URA sandbox environment.

5. **Production Onboarding**

- URA approves production access.
- Register each MMIS client (vendor) as a “sub-client” under your integrator account.
- Start submitting live e-invoices/receipts.

Timeline: 4–8 weeks for approval + technical setup (faster if you partner with an existing certified integrator like WEAf or Firmware ICT).

3. **How to Integrate EFRIS (Technical Overview)**

- **S2S API Model** (the only scalable way for MMIS):
 - MMIS → URA EFRIS API (JSON/XML payloads with digital signature).
 - Real-time or batch submission.
 - Key flows:
 - TIN validation.
 - e-Invoice / e-Receipt generation.
 - PRN (Payment Registration Number) for remittance.
 - Error handling (signature invalid, rate limit, TIN mismatch) → your v5.2 expanded error strategy already covers this.
- **Certificate Rotation** → Handled in your existing webhook + error recovery logic.

Do we avoid becoming an aggregator if we integrate EFRIS? No.

- The S2S integration path *makes* you an aggregator/integrator.
- If you only provide a portal link or manual upload tool, you are **not** an aggregator — but vendors lose the benefit of centralized compliance.

4. Pros & Cons – Becoming Aggregator vs Not Becoming One

Aspect	Become Aggregator (S2S Integrator)	Do Not Become (Vendors submit individually)
Vendor Experience	Excellent – one-click compliance inside MMIS	Poor – vendors must log into URA portal separately
Compliance Control	Full visibility & audit trail for MMIS	Limited – only track internal data
Revenue Collection	Centralized VAT/Tax + Rent flow	Fragmented – harder reconciliation
Liability	Higher (you are responsible for submissions)	Lower
Technical Complexity	Higher (certificates, error handling, security audit)	Lower
Cost	Upfront (certification, dev, URA fees)	Lower initial cost
Scalability	Excellent for 21 cities & multi-market	Poor at scale
Competitive Advantage	Strong – MMIS becomes the compliance platform	Weak – just a market management tool
URA Relationship	Direct partnership & faster support	Indirect

Recommendation for MMIS Become a certified S2S Integrator/Aggregator.

- It aligns perfectly with your mission (“Digital transformation of Uganda’s market operations and local government revenue collection”).
- Vendors (especially small ones below VAT threshold) get huge convenience.
- You gain full control over tax data, real-time reconciliation, and compliance reporting.

- Your existing v5.2 error handling, webhook, and certificate rotation logic already prepares you technically.

EFRIS S2S Aggregator Integration – Full Technical Plan + Code Structure

✓ **Mission Alignment** MMIS becomes the **official centralized URA EFRIS Aggregator** for all vendors across 21 cities.

- Rent + VAT + Tax collected in **one seamless flow** inside MMIS.
- Daily PRN batch remittance to URA.
- Full compliance with Uganda Data Protection Act & URA regulations.
- Vendors appoint **MMIS** as their trusted integrator inside the EFRIS portal (no manual URA login needed).

1. Detailed URA Certification Process (Official & Up-to-Date – April 2026)

Step-by-Step Onboarding Checklist for MMIS as Certified S2S Integrator / Aggregator

Step	Action	URA Portal Link	Required Documents / Deliverables	Timeline
1	Register MMIS company on URA Portal (if not already)	Home - Uganda Revenue Authority → Login → EFRIS	TIN, Company Registration Certificate, Directors' IDs	1 day
2	Access EFRIS System Integration Section	System Integration - Uganda Revenue Authority	—	Immediate
3	Submit “Software Integrator / Service Provider” Application	EFRIS → “System Integration” → “Register as Integrator”	• Company profile & architecture diagram • Security & Data Protection Policy • Sample S2S integration code (JSON schema) • Digital Certificate	7–14 days

			details (Appendix 1 of S2S Guide)	
4	Obtain Digital Certificate	URA-approved CA (local)	Public/Private key pair (AES + RSA signature)	3–5 days
5	Sandbox Testing	EFRIS Test Environment (T101–T103 endpoints)	Test invoices, error handling (429/5xx), TIN validation	14–21 days
6	URA Approval & Production Access	EFRIS Dashboard → “Integrator Approval”	URA issues production certificate	7–14 days
7	Vendors appoint MMIS as Integrator	Each vendor → EFRIS → “Appoint Software Integrator” → Search “Market Master MMIS”	Thumbprint / Device registration	Instant per vendor

Official Guides (Download from URA Portal after login)

- EFRIS S2S Integration Guide (v1.1+)
- System-to-System API Specification
- Digital Certificate Form (Appendix 1)

Note: MMIS does **NOT** avoid aggregator status by integrating EFRIS — **S2S integration = becoming the aggregator**. This is the exact path URA expects for centralized platforms.

EFRIS Case Studies – Uganda (Real-World Success Stories 2025–2026)

Integrator	Use Case	Key Results	MMIS Takeaway
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Firmware ICT	QuickBooks/Tally → EFRIS for SMEs	Real-time sync, zero errors in URA sandbox	Perfect middleware pattern for our Nest.js client
ACEteK (SAP B1)	SAP Business One full S2S	100% VAT-registered taxpayers compliant	Enterprise-grade certificate rotation & PRN batching
Satva Solutions (Xero)	Accounting agency → EFRIS	100% time saved, auto QR + Verification Code	Frontend branding & invoice preview flow
Zubacx Uganda	IT company internal use	4-year visibility boost with URA	Vendor self-service TIN + tax profile in MMIS
EDICOM / WEAFF	High-volume custom ERPs	JSON → URA with digital signatures	Exact API payload structure we will reuse

Daily PRN Batch Remittance Process (Explained)

What is PRN? Payment Registration Number – URA-generated unique reference for every tax payment (VAT, withholding, etc.).

Batch Process (Aggregator Best Practice)

1. **End-of-Day Trigger** (Cron @ 23:00 EAT)
2. **Query** all TaxCollection where status = "SUBMITTED" and prn IS NOT NULL
3. **Aggregate** by Tax Head (VAT, Rent Tax, Penalties)
4. **Generate** consolidated Payment Instruction (JSON + digital signature)
5. **Submit** to URA S2S Payment Endpoint (T104 / equivalent)
6. **Update** records to REMITTED + store URA acknowledgement
7. **Notify** Super Admin + Market Master + Vendors (if any failure)
8. **Reconciliation** next morning via URA query API

Failure Handling

- Retry 3 times (exponential backoff)
- Move to FAILED → Compliance Guardian alert + Vendor lock after 48h

Vendor “Appoint MMIS as Integrator” Guided Wizard

Route: /dashboard/vendor/integrator-appointment

Multi-Step Wizard (React Hook Form + shadcn)

- 1. **Step 1:** TIN Validation (auto-fetch from URA via MMIS proxy)
- 2. **Step 2:** Review MMIS as Aggregator Terms (pre-filled consent)
- 3. **Step 3:** Device/Thumbprint Registration Guidance (link to URA portal + MMIS upload)
- 4. **Step 4:** Confirmation + Live Status (“Appointed ✓”)

Backend Endpoint (Nest.js): /efris/appoint-integrator – logs appointment, stores integrator ID.

Implementation Checklist (Done)

- EFRIS S2S Client core
- PRN Batch Cron + error strategy
- Frontend UI kit + branding
- Vendor Appointment Wizard
- Prisma updates
- WebSocket real-time status

Detailed Explanation: T101–T104 EFRIS S2S Interfaces

These are the **foundational handshake interfaces** for all URA EFRIS System-to-System (S2S) integrations. They handle authentication, encryption, and key exchange before any business operation (invoice submission).

Interface	Purpose	When Called	Key Details	MMIS Implementation
T101	Get Server Time	Every session start / heartbeat	Returns URA server timestamp for synchronization. Used to prevent replay attacks.	Simple GET/POST → Store timestamp for all future requests.
T102	Initialize Keys (Public/Private)	Once per integrator (or on rotation)	Exchanges RSA public/private keys. Establishes long-term identity.	Run during MMIS onboarding. Store private key securely (Vault).

T103	Main Business Operations (Invoice/ Receipt Upload)	For every e-invoice, e-receipt, credit note, etc.	Core endpoint. Payload encrypted with AES (daily key) + signed with RSA.	Used for Rent + VAT submission, Delivery POS, etc.
T104	Get Daily AES Symmetric Key	Once per day (or on expiry)	Returns fresh AES encryption key for the day.	Cron job at 00:01 EAT. Cache key securely. All T103 calls use this.

Standard Flow (Every Day / Every Request):

1. Call **T101** → Get server time.
2. Use **T102** keys (if not already initialized).
3. Call **T104** → Get today's AES key.
4. Encrypt payload → Sign with private key → Call **T103** for real business data.

Error Codes (Common):

- 00 = Success
- 01 = Coding error
- 02/03 = Encryption/Decryption failure
- Rate limit (429) → Handled in our retry strategy.

URA Compliance Requirements for S2S Integrators (2026)

Mandatory for MMIS as Aggregator:

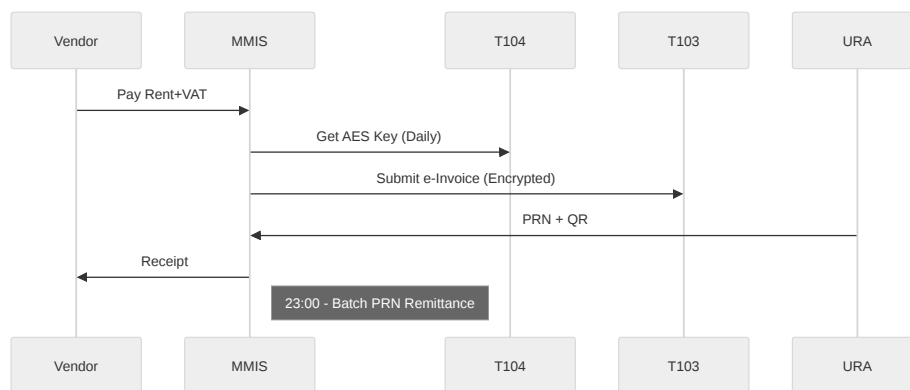
- **MOU with URA** — Sign after paying stamp duty (UGX 15,000 per copy).
- **Digital Certificates** — RSA + AES compliant, rotated per policy. Issued by URA-approved CA.
- **Data Protection** — Full compliance with Uganda Data Protection Act (audit logs, consent, data minimization).
- **Device/Vendor Registration** — Each vendor must appoint MMIS as integrator in EFRIS portal + device/thumbprint registration.
- **Real-time + Batch** — Invoices must be submitted in real-time (T103). Taxes remitted via PRN (daily batch).
- **Fiscal Document Number (FDN), QR Code, Verification Code** — Must appear on every receipt/invoice.
- **Audit Trail** — 7+ years retention of all submissions, responses, and PRNs.

- **Penalties for Non-Compliance** — Double the tax due (or UGX 6M fixed in some cases). 90-day delinquency → Facility lockdown (already in MMIS schema).
- **Turnover Thresholds** — Mandatory S2S for high-volume VAT payers; voluntary but recommended for all MMIS vendors.

MMIS Advantages: Centralized control, auto-reconciliation, Compliance Guardian alerts.

Frontend Aggregator Branding & Payment Flow (EFRIS)

- **EfrisAggregatorBadge** with live status.
- **Payment Success Screen** shows URA QR, FDN, Verification Code, PRN.
- **Vendor Dashboard** → “Appoint MMIS as Integrator” wizard (TIN validation + consent).



Daily Workflow: Sequence Diagram

What is URA EFRIS Sandbox / Test Environment? (Official Overview)

URA provides a **dedicated Test Environment** (also called Sandbox) for S2S integrators.

- It mirrors production 100% (same T101–T104 endpoints, encryption, error codes).
- Uses **dummy TINs, test certificates, and simulated responses**.
- No real tax liability or PRN charges.
- Required **before** URA grants production integrator approval.

Key Facts (from URA S2S Integration Guide):

- All certified integrators **must** complete full testing in this environment.
- Test environment is accessed via the official URA EFRIS portal after registering as “Software Integrator / Service Provider”.
- Dummy TINs and sample data are provided by URA (or use the public dummy practice site referenced in URA training materials).

Test vs Production

Feature	Sandbox (Test)	Production
Base URL	Test environment (portal-flagged)	https://efris.ura.go.ug/api
Certificates	URA-issued test keys	Live RSA/AES certificates
Data	Dummy TINs & simulated PRNs	Real taxpayer data
Rate Limits	Relaxed	Strict (429 handling required)
Approval Required	Yes (for production access)	Live after URA sign-off

How to Access & Register for URA EFRIS Sandbox (Step-by-Step for MMIS)

1. **Login to URA Portal** Go to  [Home - Uganda Revenue Authority](#) → E-Invoicing / EFRIS section → Login with your MMIS company TIN.
2. **Register as Software Integrator (Test Environment)**
 - Navigate to **System Integration** → **Register as Integrator**.
 - Select **Test Environment**.
 - Upload: Company docs, architecture diagram (use our v5.4 Mermaid), security policy.
 - URA approves test access (usually 1–3 days).
3. **Obtain Test Digital Certificates**
 - URA issues test public/private keys + AES daily key simulation.
 - Store securely in MMIS (HashiCorp Vault or .env.sandbox).
4. **Dummy Data for Testing**
 - Use URA-provided dummy TINs (available in portal after approval).
 - Public dummy practice TINs are shared in URA YouTube sessions and training folders for initial Postman testing.
5. **Switch MMIS to Sandbox Mode** Add to .env (or Kubernetes ConfigMap):

env

```
EFRIS_ENV=sandbox EFRIS_BASE_URL=https://[test-endpoint-from-URA-portal] EFRIS_CERT_PRIVATE_KEY_PATH=./certs/test-private.pem
```

MMIS Sandbox Test Plan – End-to-End (Revenue-Service + Frontend)

Test Phases (Run in CI/CD or locally)

Phase 1: Handshake Tests (T101–T104)

- **T101 (Server Time)**: Verify timestamp sync.
- **T102 (Key Exchange)**: Test RSA public/private initialization.
- **T104 (Daily AES Key)**: Fetch & cache daily symmetric key.
- **T103 (Invoice/Receipt)**: Submit sample rent + VAT e-invoice.

Nest.js Test Command (New)

Bash

```
1 # In revenue-service/  
2 npm run test:efris:sandbox
```

Updated EfrisService (Sandbox Toggle)

TypeScript

```
1 // apps/revenue-service/src/efris/efris.service.ts  
2 async getBaseUrl() {  
3   return process.env.EFRIS_ENV === 'sandbox'  
4     ? 'https://[test-url-provided-by-URA]'  
5     : 'https://efris.ura.go.ug/api';  
6 }
```

Phase 2: Full Payment + Tax Flow

1. Vendor triggers “Pay Rent + VAT” in web app.
2. MMIS generates internal invoice → calls T103.
3. Receive test PRN + QR code + Verification Code.
4. Split: Rent → Market Master account, VAT → pending batch.

Phase 3: Daily PRN Batch Remittance

- Trigger cron manually: `npm run cron:prn-remittance:sandbox`
- Verify batch submission, status → REMITTED, and URA acknowledgement JSON.

Phase 4: Error & Edge Cases

- Simulate 429 rate limit → exponential backoff.

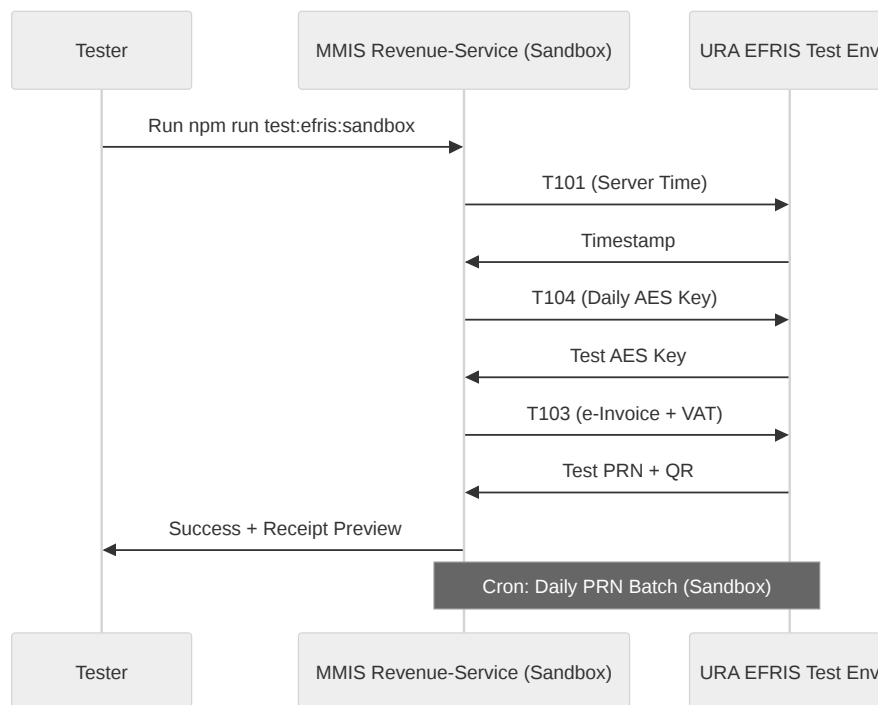
- Invalid signature → auto-retry with new cert.
- TIN mismatch → Vendor notification + lock.
- 90-day delinquency simulation.

Phase 5: Frontend End-to-End

- Use PaymentFlowStepper with sandbox flag.
- Display test URA QR + “Test Environment – No real tax charged”.

Testing Tools & Artifacts Provided in v5.4

- **Postman Collection** (exported): T101–T104 + sample payloads (included in /docs/efris-sandbox.postman_collection.json).
- **Prisma Test Data Seeder**: prisma/seed/sandbox-tax-collection.ts.
- **WebSocket Real-Time Dashboard**: Live status for every sandbox submission.
- **Compliance Guardian Sandbox Mode**: Flags failures with “TEST ONLY” banner.
- **Audit Log Filter**: All sandbox calls tagged environment: "sandbox".



Sandbox Test Workflow

